

Anant Kumar Gupta

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EDUCATION

University of Edinburgh

Bachelor of Engineering in Computer Science

- Graduated with First Class Honours.
- Relevant Courses: Advanced Algorithms & Data Structures, Operating Systems, Computer Security, Natural Language Processing, Data Science, Computer Systems, Object Oriented Programming.

Edinburgh, United Kingdom

September 2021 – May 2025

WORK & LEADERSHIP EXPERIENCE

Marshall Wace – global hedge fund managing systematic and discretionary strategies.

Software Engineer, Graduate Rotational Programme

- Rotate across engineering teams on four-month cycles, spanning the AI platform, infrastructure, quant pricing and investment tooling.
- Build and maintain internal tooling used daily by investment and research teams, primarily in Python, Java and C++.
- Own features end to end, from design and peer code review through deployment and ongoing support.

London, United Kingdom

September 2025 – Present

University of Edinburgh, School of Informatics

Tutor & Lab Demonstrator

- Tutored Introduction to Computation, the first-year Haskell course, running weekly tutorials, marking coursework and giving written feedback to students over three years.
- Demonstrated large lab sessions for Introduction to Object Oriented Programming, debugging student code and resolving environment and installation issues.

Edinburgh, United Kingdom

September 2022 – May 2025

J.P. Morgan

Software Engineering Intern

- Developed a Symphony chatbot for traders, integrating Python and Java microservices secured with OAuth, enabling traders to retrieve trade data using natural language queries.
- Designed and implemented Kubernetes deployment and network policies for secure, reliable rollout.
- Wrote unit tests in Java and Python with Mockito and JUnit to over 70% coverage, and made the chatbot reusable by other teams in the division by swapping a single Java microservice.

Glasgow, United Kingdom

June 2024 – August 2024

Institute for Computing Systems Architecture

Research Intern, School of Informatics

- Developed a domain-specific language for describing processors in Lean4 under Prof. Vijay Nagarajan and two research associates.
- Constructed a five-stage pipelined MIPS processor with the DSL, testing for bugs and proposing additions to the language for smoother simulation.
- Led refactoring and documentation, and created a continuous integration pipeline on GitHub.

Edinburgh, United Kingdom

February 2023 – July 2023

Centrum Wealth Ltd. – wealth management and investment firm in India.

Analyst Intern

- Produced more than fifteen client portfolio reports with advanced Excel tooling, setting out portfolio performance over defined periods.
- Analysed and compared Alternative Investment Funds on historical performance, risk metrics and investment strategy to inform client decisions.

New Delhi, India

July 2022 – September 2022

PROJECTS

Recipe Search Engine (Group Project)

Search engine over a scraped recipe corpus: Flask and MongoDB backend with NLTK tokenisation and stemming, marisa-trie query autocomplete and a JavaScript front end, containerised with Docker.

Optiver Trading Bot (Hackathon)

Second place at the HackTheBurg hackathon with a pair-trading bot exploiting market inefficiencies between an ETF and its constituent components.

AutoFold (Assistive Robot)

Automated clothes-folding system built with Raspberry Pi, Pico and stepper motors to assist individuals with mobility limitations.

Airbnb Edinburgh (Data Science)

Analysis of Edinburgh Airbnb listing prices using data cleaning, PCA and visualisation with Matplotlib and Seaborn, collaborating with a team over git.

SKILLS, ACTIVITIES & INTERESTS

Technical Skills: Proficient – Python, Java, Haskell; Intermediate – C/C++, SQL; Basic – C#, JavaScript, CSS, React.

Certifications & Training: CS50: Introduction to Computer Science; CS50: Artificial Intelligence with Python.

Activities: Head of Alternatives Sector, Edinburgh University Trading & Investment Club – UK's largest student-led fund (£89,000 AUM); won "Best Pitch" and "Best Report" among 50+ submissions. InfPals Leader; Director of Marketing, Edinburgh University Indian Society; Head of Informatics and Mathematics, ESRA.

Interests: Painting, flute, machine learning, investment portfolio management, filmmaking.

Languages: Fluent in English and Hindi.